AI Research Agent

Problem Statement, Intended Architecture, Research Workflow, and Verification Framework

1. Problem Statement

The purpose of the AI Research Agent is to automate a time-consuming research process. When a new technical problem or project idea is identified, substantial research is required to understand what has already been built and published. Existing GitHub projects, Hugging Face repositories, and research papers need to be discovered, studied, compared, and summarized before starting implementation. Doing this manually can take many days or even months depending on the scope of the research.

The main goal is therefore to reduce the time and effort required for prior-art and research analysis while producing a structured understanding of existing work. The system should help avoid unnecessarily rebuilding something that already exists and should help identify what is different, missing, or potentially useful for the proposed project.

2. Intended User Flow

Step	What happens
1	User provides an initial problem statement.
2	Orchestrator asks follow-up questions until the project/problem is clearly understood.
3	Orchestrator produces a clarified problem statement and research requirements.
4	The clarified problem statement is provided to the research agents.
5	Existing-project research searches relevant GitHub and Hugging Face work.
6	Research-paper research finds directly or closely relevant academic papers.
7	Evaluation agent checks the discovered evidence against explicit criteria.
8	Orchestrator combines the verified findings.
9	System produces a structured final research report.

3. Orchestrator Agent

The Orchestrator is the manager of the research workflow. It receives the user's initial problem statement and asks follow-up questions until it has a sufficiently clear understanding of the proposed project.

The Orchestrator should clarify the problem, expected input and output, important constraints, datasets or technologies involved, and other requirements that affect the research. It should then create a clear research specification that the other agents can use.

The Orchestrator should coordinate the workflow rather than performing every research task itself.

4. Agent 1 — Existing Project Research Agent

The first research agent investigates existing implementations and projects related to the clarified problem statement. Its primary sources include GitHub repositories and Hugging Face repositories or related implementation resources.

For each relevant project, the agent should try to capture structured information such as:

• What problem the project solves

• Input and output

• Dataset or data source

• Model and architecture

• Method or approach

• Evaluation and reported results

• Repository and implementation details

• Reproducibility information

• Known limitations

The purpose of this agent is to answer: “What has already been built?”

5. Agent 2 — Research Paper Agent

The second research agent searches for academic research related to the clarified problem. It should identify papers that either directly match the problem or are closely relevant and provide useful methods, datasets, architectures, experiments, or insights.

The goal is not to restrict the search only to papers with identical wording. Closely related research may reveal techniques or approaches that can be adapted to the proposed project.

The purpose of this agent is to answer: “What has already been researched?”

6. Agent 3 — Evaluation / Verification Agent

The Evaluation Agent is responsible for checking the discovered projects and papers against explicit criteria. It should not merely perform keyword matching or give a vague statement such as “this paper is good.” It should provide evidence for its judgments.

Criterion	Evaluation question
Problem relevance	Does the work address the same core problem?
Input compatibility	Does it use compatible input data or information?
Output compatibility	Does it produce the type of output required?
Dataset relevance	Is the dataset relevant to the proposed problem?
Method relevance	Is the method applicable to the proposed approach?
Model relevance	Is the model/architecture relevant?
Experimental evidence	Does the work demonstrate the method experimentally?
Reproducibility	Is there enough information/code/data to reproduce or study it?
Research gap	What remains unresolved or different from the proposed problem?

7. Verification Philosophy

The research system should distinguish between finding information and verifying information. A research agent can discover a paper or repository, but discovery alone does not prove that the work is relevant to the user's problem.

For each important judgment, the system should define a question, explicit criteria, and evidence that supports the resulting judgment. This follows the webinar framework discussed during the project: define what is being judged, define the question, and define the criteria used to judge it.

The evaluator should therefore produce evidence-backed results such as verified, partially verified, not verified, or unknown where appropriate, rather than relying only on a model's unsupported confidence.

8. Recommended Architecture

User → **Orchestrator** → **Clarified Problem Statement** → **Research Agents** → **Evidence** → **Evaluation/Verification** → **Orchestrator** → **Final Research Report**

The existing-project research and paper research can operate in parallel because both can start from the same clarified problem statement. Their findings can then be passed to the evaluator.

9. Final Report

• Original and clarified problem statement

• Research requirements and constraints

• Relevant existing GitHub/Hugging Face projects

• Relevant research papers

• Methods and architectures found

• Datasets and experimental setups

• Criterion-by-criterion evaluation

• Evidence supporting important judgments

• Limitations of existing work

• Potential research gaps or differences

• Structured summary for deciding the next research/implementation step

10. Development Order

1. Build the Orchestrator and clarification flow.

2. Build the existing-project research agent.

3. Build the research-paper agent.

4. Define the evaluation criteria before implementing the evaluator.

5. Build the Evaluation/Verification Agent.

6. Connect research evidence to the evaluator.

7. Return verified findings to the Orchestrator.

8. Generate the final structured research report.

9. Add human review for cases where the criteria cannot reliably establish correctness.

11. Core Design Principle

The goal is not to build many AI agents for the sake of having agents. The goal is to divide the research workflow into responsibilities that can be clearly executed and, where possible, verified. Research agents discover information; the evaluator checks evidence against explicit criteria; the Orchestrator coordinates the process and produces the final report.

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